

Ivan Kuang

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EDUCATION

University of California, Berkeley

Expected Graduation - **May 2027**

B.S. Mechanical Engineering Undergraduate

Relevant Coursework: SolidWorks Modeling and FEA, Solid Mechanics, Thermodynamics, Materials Behavior, Electronics for IoT, MATLAB, Python, Machine Learning, Manufacturing / GD&T

SKILLS

Design & Simulation: SolidWorks (CAD/FEA), ANSYS, OnShape, Fusion 360, SketchUp

Programming & Data: Python, MATLAB, Jupyter Notebook, Data Analysis/Machine Learning

Manufacturing: 3D Printing (FDM/Binder Jetting), TIG/MIG Welding, Woodworking, CNC Machining, GD&T

Awards/Certifications: ASME 2025 Cadathon GOLD, SkillsUSA Cabinetmaking Regional GOLD, OSHA 10 Hour Certified

EXPERIENCE

Undergraduate Researcher | Gu Research Group

November 2025 - Present

- Developed a custom Binder Jetting system from scratch, achieving a near 90% cost reduction over commercial alternatives by creating a scalable R&D platform designed to rigorously test and optimize the structural integrity of printed green bodies.
- Resolved chronic gypsum powder leakage within high-tolerance aluminum piston assemblies by designing, machining, and integrating custom-fit O-ring seals, ultimately ensuring long-term operational reliability and system cleanliness.
- Significantly accelerated the overall R&D cycle by developing a real-time CNN-based defect detection system utilizing image segmentation, successfully automating visual inspection protocols to reduce manual data analysis time.

Occupancy Cell Engineer | Cal Solar Racing

September 2024 - Present

- Designed the GEN XI Occupancy Cell using SOLIDWORKS by synthesizing ergonomic jig data with complex race structural regulations, including suspension hardpoints, steering geometry, and regulatory ballast integration, while utilizing a top-down CAD methodology and composite sandwich panel analysis to optimize driver safety and overall vehicle performance.
- Conducted comprehensive structural analysis in Ansys to validate occupancy cell structure with WSC impact standards, correlating simulation predictions with physical 3-point bend tests on composite panels to verify material performance under specified load cases.
- Developed a manufacturing strategy for 12+ composite panels, creating custom assembly jigs and nesting patterns with precise GD&T specifications to significantly minimize carbon fiber waste and streamline final structural integration.

Airframe Engineer | Space Technology and Rocketry

August 2025 - January 2026

- Designed LE3 rocket metal runners and optimized support spacing via SolidWorks FEA to ensure the structural integrity of critical fuel lines during 10 m/s impacts, a design limit derived from the terminal descent velocity calculated post-drogue parachute deployment.
- Engineered a 12-inch airframe extension to eliminate critical high-stress couplers, reducing the total chassis weight by 5% while maintaining a verified 2+ Factor of Safety under anticipated launch load simulations.

Research Internship | Caltech Seismo Lab

March 2023 - August 2023

- Collaborated within a four-person engineering team to develop a cost-effective alternative to professional seismic arrays, establishing a robust regression model to detect local seismic events up to magnitude 4.0 using low-cost sensor hardware.
- Executed advanced signal processing by converting raw data from Raspberry Shake sensors into calibrated displacement metrics, utilizing ObSpy to perform precise noise reduction and signal filtering, ultimately enabling accurate calculation of local seismic magnitude on the Richter scale.

PROJECTS

Power Wind Turbine

August 2024- December 2024

- Designed and FDM 3D-printed a wind turbine system using PLA, using FEA to optimize a 178g tower that sustained a 3.5 kg static load and achieved a 500 mW peak power output through iterative blade design.

Combat Robotics: Skofnung

August 2025- January 2026

- Prototyped a non-traditional crank and slider shuffle drive for a 1lb combat robot, using Onshape CAD to integrate custom drive modules with a high-mass weapon assembly and utilizing SLA-printed 40A flexible resin feet to enhance drive torque and traction.

C.A.R.E. Design

April 2025

- Designed ISO Class 1–5 cleanroom Autonomous mobile robot featuring a 3-DOF tray system designed to minimize transport vibrations for 5–50 lb semiconductor payloads.
- Incorporated various subsystem CAD assemblies integrating a Mecanum drivebase and a stainless steel robotic arm optimized for high-precision pick-and-place trajectories.